Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-3 | Winter-2024

Course Code	:	2305CS303	**Date**	:	18-10-2024
Course Name	:	Python for Data Science	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Enumerate the list and its methods with example.	**4**
	(B)	Differentiate between the tuple and sets in python.	**3**
		OR	
		Differentiate between the List and sets in python.	
	(C)	What is data type? List out the types of data types with example.	**7**
		OR	
		Explain variable with suitable example.	
Q.2	**(A)**	Write a program to print the fibonacci series upto n_terms using Recursion.	**4**
	(B)	Explain For loop statement with any example.	**3**
		OR	
		Explain While loop statement with any example.	
	(C)	Explain Types of Python Function Arguments with any example.	**7**
		OR	
		Write a program to use of nested if statement.	
Q.3	**(A)**	Explain Type of Indexing in NumPy.	**4**
	(B)	Write an out of the following code.	**3**

```
import numpy as np
a = np.array([5, 72, 13, 100])
b = np.array([2, 5, 10, 30])

add_ans = a+b
print(add_ans)

add_ans = np.add(a, b)
print(add_ans)

c = np.array([1, 2, 3, 4])
```

```
add_ans = a+b+c
print(add_ans)

add_ans = np.add(a, b, c)
print(add_ans)
```

OR

Explain any two NumPy Arithmetic Operators.

(C) List any 5 Array creation function with description. **7**

OR

List out any 5 NumPy data types with description.

Q.4 **(A)** Create a DataFrame by passing a dictionary of Series objects: **4**

```
import pandas as pd

people_dict = {
    "weight": pd.Series([68, 83, 112], index=["alice", "bob", "charles"]),
    "birthyear": pd.Series([1984, 1985, 1992], index=["bob", "alice", "charles"],
name="year"),
    "children": pd.Series([0, 3], index=["charles", "bob"]),
    "hobby": pd.Series(["Biking", "Dancing"], index=["alice", "bob"]),
}

people = pd.DataFrame(people_dict)
people
```

(B) What is Reset_Index() in Pandas? **3**

OR

Explain use of Reset_Index() method in pandas.

(C) List out any 5 pandas series function with description. **7**

OR

Explain Concatenation and Merge with example.

Q.5 **(A)** Draw a graph for following code **4**

```
import matplotlib.pyplot as plt
import numpy as np
# Define X and Y variable data
x = np.array([1, 2, 3, 4])
y = x*2
plt.plot(x, y)
plt.xlabel("X-axis")  # add X-axis label
plt.ylabel("Y-axis")  # add Y-axis label
plt.title("Any suitable title")  # add title
plt.show()
```

(B) Draw a graph for following code **3**

```
# importing package
import matplotlib.pyplot as plt
# create data
x = [10,20,30,40,50]
y = [30,30,30,30,30]
# plot line
plt.plot(x, y)
plt.show()
```

OR

Draw a graph for following code

```
# importing package
import matplotlib.pyplot as plt
# create data
x = [10,20,30,40,50]
y = [30,30,30,30,30]
# plot lines
plt.plot(x, y, label = "line 1")
plt.plot(y, x, label = "line 2")
plt.legend()
plt.show()
```

(C) Write a code and draw a line in a diagram from position (1, 3) to (2, 8) then to (6, 1) and finally to position (8, 10): **7**

OR

Explain type of Types of Charts in Matplotlib

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